

Business Requirements

This document defines the business requirements for the Release Quality Analytics Platform project.

Project Background

Software release processes generate large amounts of testing and defect data. This project demonstrates an end-to-end analytics workflow that transforms raw QA data into meaningful business insights using SQL, Python and Power BI.

Business Problem

- Manual reporting is time-consuming.
- Release visibility is limited.
- KPI calculation is manual.
- There is no centralized reporting solution.

Business Goal

Develop a centralized Release Quality Analytics Platform capable of:

- Monitoring release quality
- Tracking software defects
- Measuring testing performance
- Supporting release readiness decisions
- Providing interactive dashboards

Stakeholders

Role	Responsibility
Product Owner	Monitor release readiness
QA Lead	Track testing progress
Project Manager	Review project health
Business Analyst	Support reporting and decision making
Software Tester	Analyze execution metrics

Epic

Release Quality Analytics Platform

User Stories

As a Product Owner

I want to monitor release quality from a single dashboard so that I can quickly determine whether a release is ready for production.

As a QA Lead

I want to analyze bug severity and status distributions so that I can prioritize testing activities.

As a Project Manager

I want to review project KPIs so that I can monitor overall software quality.

As a Test Lead

I want to compare tester execution performance so that workload distribution can be

evaluated.

Acceptance Criteria

- Display release KPIs
- Show bug severity distribution
- Show bug status distribution
- Display tester performance
- Calculate KPIs using DAX
- Perform SQL-based aggregations
- Include Python-generated analytical datasets
- Support interactive Power BI filtering

Expected Business Value

- Reduce manual reporting effort
- Improve release visibility
- Support data-driven decision making
- Demonstrate an end-to-end analytics workflow